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DATA DESCRIPTOR

SeizelT2: Wearable Dataset Of Patients With Focal Epilepsy

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The increasing technological advancements towards miniaturized physiological measuring devices have enabled continuous monitoring of epileptic patients outside of specialized environments. The large amounts of data that can be recorded with such devices hold significant potential for developing automated seizure detection frameworks. In this work, we present SeizelT2, the first open dataset of wearable data recorded in patients with focal epilepsy. The dataset comprises more than 11,000 hours of multimodal data, including behind-the-ear electroencephalography, electrocardiography, electromyography and movement (accelerometer and gyroscope) data. The dataset contains 883 focal seizures recorded from 125 patients across five different European Epileptic Monitoring Centers. We present a suggestive training/validation split to propel the development of AI methodologies for seizure detection, as well as two benchmark approaches and evaluation metrics. The dataset can be accessed on OpenNeuro and is stored in Brain Imaging Data Structure (BIDS) format.

Background & Summary

Epilepsy is one of the most common neurological disorders, affecting around 1% of the global population¹, and seizures are its primary clinical manifestation². Seizures are transient phenomena resulting from the activation and distribution of abnormal or excessive neuronal activity within limited or diffuse brain networks³. The signs observed by others and the symptoms reported by the patient depend on the brain region(s) where the abnormal activity arises and its further propagation pattern⁴. Therefore, patients can present with different manifestations, including but not restricted to altered consciousness, abnormal visual, behavioral, auditory, sensorimotor, psychic, and autonomic phenomena, or a combination of these^{4,5}. The International League Against Epilepsy (ILAE) classifies seizures into three main groups based on their onset: focal, when the network is limited to one hemisphere; generalized, defined by the “rapid engaging of bilaterally distributed networks”; and unknown if the onset cannot be identified⁵. The unpredictability of epileptic seizures has a negative impact on the quality of life in many patients⁶, and epilepsy is associated with an increased risk for accidents and mortality^{7,8}.

After the diagnosis of epilepsy, patients receive pharmacological treatments for seizure control. In some drug-resistant cases, additional dietary or surgical treatments may be beneficial. In outpatient settings, documenting seizures is of great importance for monitoring the course of the disease and adjusting the treatments. Currently, clinicians rely on the seizures reported by the patient in diaries. However, seizure diaries have shown a documentation sensitivity of less than 50%^{9,10}.

Furthermore, 30% of patients with epilepsy have drug-resistant epilepsy¹¹. In many of these cases, a comprehensive evaluation is done, including recording and analysing seizures using video electroencephalography (vEEG). As vEEG is a costly and time-consuming examination, patient selection is crucial for effective use of resources. Due to the unpredictability of seizures, it is uncertain whether a seizure will be captured during a vEEG admission, reducing the diagnostic yield and delaying clinical decision-making^{12,13}. Therefore, clinicians require more efficient methods to improve patient monitoring.

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The development of small wearable recording devices has surged in recent years. These devices can measure physiological data for long periods outside the hospital, enabling continuous monitoring of patients¹⁴. Wearable EEG technologies are often designed to discreetly encapsulate a reduced set of electrodes intended to measure atypical regions of the scalp, such as behind-the-ear. Devices like the Sensor Dot (SD), from Byteflies (Antwerp, Belgium)¹⁵, were developed for the purpose of monitoring epilepsy patients during their daily lives. The development of the SD device was accomplished within the SeizeIT1 study¹⁶, where patients with focal epilepsy were monitored in-hospital with vEEG, together with an additional set of electrodes placed behind-the-ear, that mimicked the wearable setup and its data characteristics. Preliminary validation studies using this recording setup showed promising results for the detection of seizures in a controlled environment with a simulated wearable behind-the-ear EEG (bte-EEG) setup^{17,18}. It was concluded that the performance between scalp and wearable EEG detection algorithms was similar in a small group of patients with temporal lobe seizures, proving the potential of using such a modality for seizure detection. Other types of wearables have been used to monitor different physiological signals, such as electrocardiography (ECG), electromyography (EMG), accelerometry (ACC), gyroscope data (GYR), electrodermal activity (EDA) and combinations of these¹⁹. When data is recorded outside of the hospital, clinicians do not have video information to assess behavioural changes caused by seizures, such as spasms, tonic or clonic episodes. By combining information from different physiological signals, using multimodal wearables, the detection of seizures can be improved^{20,21}.

The use of wearable devices as a longitudinal monitoring tool presents a significant challenge in analyzing and annotating seizures due to the vast amounts of collected data. Clinicians are not typically trained to identify seizures in non-standard EEG montages. Additionally, it is not feasible to manually annotate thousands of hours of data from multiple patients. It is of high interest to develop automated methods for detecting seizures based on the recorded data. Significant work has been done on automated machine learning (ML) frameworks to detect seizures based on changes in many types of recording modalities, such as scalp-EEG^{22,23}, ECG²⁴ and EMG²⁵. However, research on wearable EEG is mainly unexplored and only initial findings have been published^{10,21,26,27}. The development of ML algorithms for detecting seizures in wearable data involves significant adaptations to the pipelines due to differences in the properties and morphology of the data compared to standard modalities (vEEG). The use of bte-EEG for detecting focal seizures in practice is limited by the current automated frameworks' performance²⁸. Despite the higher sensitivities achieved, when compared to seizure diaries, the number of false alarms is relatively high. The main ingredient of such automated methods is data. In literature, the majority of public datasets recorded from patients with epilepsy contain only full-scalp EEG data²⁹. There is a limited amount of open EEG datasets, and, to our knowledge, there are no publicly available datasets with recordings containing data from wearable devices from patients with epilepsy. The many published studies on ML for seizure detection are trained and tested on a heterogeneous cohort and use various validation methods. The training and test data are measured with different equipment, pruned in diverse ways and might contain non-continuous measurements, impeding generalization capabilities and decreasing the robustness of the seizure detection algorithms. The validation methods can vary depending on how the data is cleaned and the reported metrics are not standardized, creating a need for sharing data used in the development of such frameworks and a common evaluation pipeline³⁰.

The dataset presented in this work was created with the objective of promoting the development of automated focal seizure detection frameworks in continuous wearable data. The study scope is to evaluate the usability and feasibility of the Sensor Dot device in the hospital environment. By recording data from the gold-standard vEEG and from the wearable device, it is possible to compare the monitoring capabilities of the Sensor Dot and develop clinical tools to aid clinicians in accurately count seizure occurrences. To the best of our knowledge, this is the first and largest phase 3 clinical study containing public multimodal (bte-EEG, ECG, EMG, ACC and GYR) wearable data recorded from patients with focal epilepsy.

Methods

Study Design and Participants. The SeizeIT2 project (clinicaltrials.gov: NCT04284072), a multicenter, prospective study, was carried out to validate the Sensor Dot device in adult and pediatric patients with epilepsy in a controlled environment. Participants were included if they had a history of refractory epilepsy and were admitted to the Epilepsy Monitoring Unit (EMU) for routine 24 hour or long-term vEEG monitoring as a presurgical evaluation procedure. The exclusion criteria included patients with skin conditions or allergies that prevented the placement of the electrodes and adhesives or had implanted devices, such as neurostimulators or pacemakers. All participants provided written informed consent. The data collection started on January 10, 2020, and ended on June 30, 2022. The study was approved by the UZ Leuven ethics committee (approval ID: S63631), anonymization and sharing of the data was also approved by the same committee (S67350 - amendment 1). During the study, participants underwent standard vEEG monitoring procedures with the Sensor Dot device used as an additional recording modality. vEEG data were analyzed by the clinicians at each center, and seizure annotations were mapped onto the wearable data using the same start and end times, as well as clinical characteristics such as seizure type, lateralization, localization, and any additional symptoms.

The dataset comprises 125 patients (51 female, 41%) from 5 different European EMUs: University Hospital Leuven (Belgium), Freiburg University Medical Center (Germany), RWTH University of Aachen (Germany), Karolinska University Hospital (Sweden) and Coimbra University Hospital (Portugal). Figure 1 shows the distribution of the number of patients recorded in each center. The University Hospital Leuven was the only center that enrolled pediatric patients. The dataset includes only data from patients with focal epilepsy who experienced one or more seizure episodes during the monitoring period. In total, eight different annotators analysed the vEEG data to obtain information regarding the seizures recorded in the dataset (seizure type, onset location and clinical symptoms). The annotators included two neurologists from the University Hospital Leuven (one for adult patients and one for pediatric), one neurologist from the Freiburg University Medical Center,

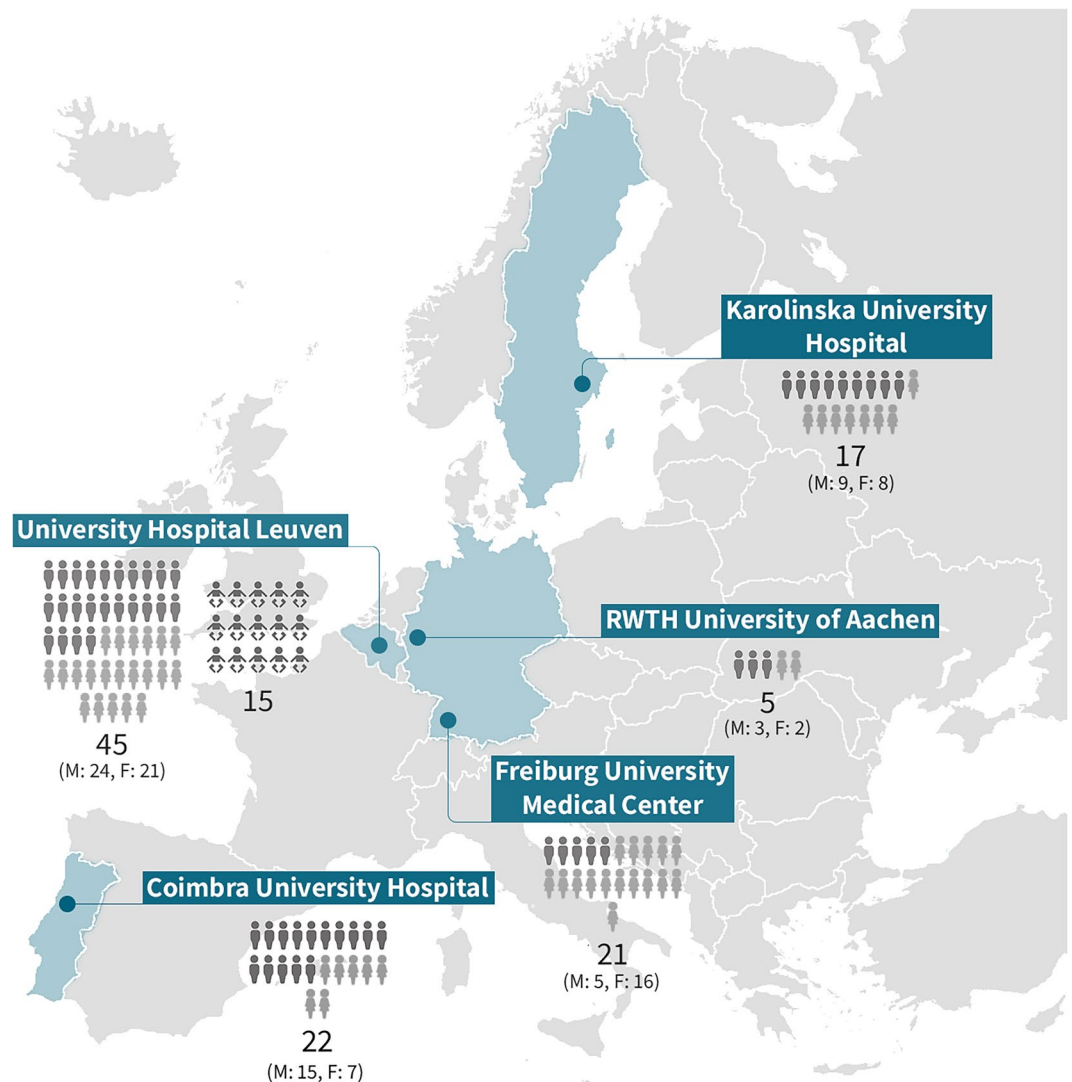


Fig. 1 Number of participants per EMU included in the SeizeIT2 (M: male, F: female). The dataset includes patients from 5 different centers across 4 different European countries.

three neurologists/epileptologists from the RWTH University of Aachen, one neurologist from the Karolinska University Hospital and one neurologist from the Coimbra University Hospital.

Recording Setup. The participants were recorded with the specific center's vEEG monitoring equipment, where the EEG electrodes were placed according to the 10-20 system or the 25-electrode array of the International Federation of Clinical Neurophysiology. The vEEG data was recorded with a sampling frequency of 256 Hz. The SD device was used to record wearable data simultaneously with the vEEG. The device has a size of 24.5 x 33.5 x 7.73 mm and weighs approximately 6.3 grams. The wearable device measures EEG, ECG and EMG data at a sampling frequency of 250 Hz and movement data at 25 Hz, and has a battery life of approximately 24 hours. The electrodes connected to the device were Ag/AgCl cup electrodes, with impedance values below 5 kΩ. Two recording devices were used: one placed in the patient's upper back using a patch and connected to electrodes attached behind the ear, on the mastoid bone (EEG SD); another placed on the left side of the chest, with two electrodes extended to the lower left rib cage and the fourth intercostal space in the left parasternal position to measure ECG, and two electrodes extended to the left deltoid muscle to measure EMG data (ECG/EMG SD). The module itself contains ACC and GYR sensors, which measure movement data. The SD setup is presented in Fig. 2. The EEG SD electrode placement depended on the patient's medical history and is based on the seizure type and onset. When the seizures were suspected to originate from the left hemisphere, two electrodes were placed on the left side and one on the right side, forming one left same-side channel and one cross-head channel. Analogously, if seizures were suspected to originate from the right hemisphere, the same-side channel was derived from two electrodes placed behind the right ear. The dataset includes patients who were suspected to have generalized seizures (but had focal seizures) as well, and in this case, the cross-head channel was non-existent and replaced by an additional lateral channel by using two electrodes on each ear. The dataset contains only data where seizures with onset on the left hemisphere were recorded in patients in which the left setup was used, and seizures originating

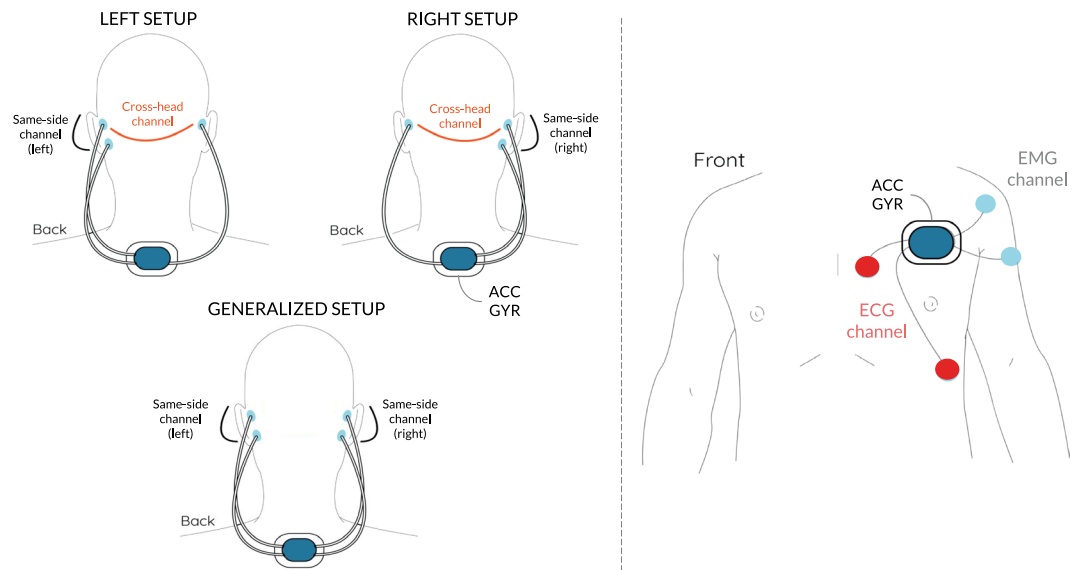


Fig. 2 Schematic of the placement of the Sensor Dot device. The bte-EEG setup on the left is separated between left, right and generalized setups (where the left and right setups contain a cross-head and a same-side channel, the generalized contains two same-side channels, left and right). The ACC and GYR data is recorded from the bte-EEG recording module and the ECG and EMG recording module. Adapted from³⁴.

from the right hemisphere were recorded with the right setup. Seizures originating from the left or right hemisphere recorded with the generalized setup are also included in the dataset. There were no seizures in which the hemisphere of origin was different than the setup lateralization. The placement and impedance of each module were checked at the beginning and routinely during the monitoring sessions.

Dataset Content. The complete dataset contains around 11 600 hours of wearable data (5490 hours of male, 5018 hours of female and 1059 hours of pediatric data), with a total of 2850 recordings with an average duration of 4 hours per recording (most recordings are sequential from 24 hour recording sessions). Four different modalities were recorded for most participants: bte-EEG, ECG, EMG and movement data. All participants' data within the dataset contain wearable bte-EEG. In 3% of the dataset, ECG, EMG and movement data were not included due to technical failures or errors in the setup. In total, 883 focal seizures were recorded with the wearable device. The mean duration of the recorded seizures was 58 seconds, ranging between 3 seconds and 16 minutes (the seizures recorded in male participants have a mean and standard deviation duration of 82 and 91 seconds respectively, for female participants 61 and 70 seconds and for pediatric participants 20 and 23 seconds). The majority of the seizures were focal aware (FA) and focal impaired awareness (FIA), with 316 and 391 occurrences, respectively. From the remaining seizures, 55 were focal-to-bilateral tonic clinic (FBTC), 115 were focal with unclear awareness status, where 17 were subclinical, and 6 had unknown or unreported onset. There was a predominance of seizures with onset on the left hemisphere (44%). In 12% of the seizures, the onset was located in the right hemisphere, 1% had a bilateral onset and in 43% of the seizures the onset was unclear. Regarding localization, the seizure onsets were distributed over the central, frontal, temporal, occipital, parietal and insula lobes, with a predominance of temporal lobe seizures (30%). Several seizures recorded could not be paired with a clear onset lobe (26%). Table 1 shows detailed numbers of seizure occurrences and their respective lateralization and localization characteristics. The supplementary file containing Table S1 includes the same information per patient, along with the patients' clinical characteristics, recordings duration, and of seizure occurrences per type.

Data Records

The dataset can be accessed at the OpenNeuro repository (<https://doi.org/10.18112/openneuro.ds005873.v1.1.0>)³¹ and conforms to the BIDS format³². The data structure in the repository is represented in Fig. 3. The file 'dataset_description.json' contains general information regarding the BIDS version used, licensing, authorship and acknowledgements. The 'events.json' file gathers categories and respective descriptions of all event types annotated in the dataset, related to every event file associated with each recording. The information about the participants' sex is stored in the 'participants.tsv' file, with the associated 'participants.json' file with a description of the sex categories. Within the dataset, each participant's data is gathered in a separate folder ('sub-xxx'). Every main folder contains one subfolder ('ses-001') and within the latter, four different subfolders are organized to store each modality's data ('eeg', 'ecg', 'emg' and 'mov', for bte-EEG, ECG, EMG and movement- ACC and GYR- data respectively). For every modality's folder, each recording contains a '.edf' and a '.json' file, named with an extension '(...)_run-XX_' as an identifier. In the 'eeg' folder, there is an additional '.tsv' file associated to each recording with the extension '_events'. The '.edf' files contain the data in European Data Format (EDF). The '.json' files include information about the sampling frequency, channel counts and placement, duration of the recordings and description of the task. The '_events.tsv' files gather the annotations of each recording, with the fields described in the 'events.json' file: start time and duration of the recording (in the fields 'dateTime' and

		FA	FIA	FBTC	Focal	Subclinical	Unknown
	Total	316	391	55	98	17	6
Lateralization	Left	170	160	26	20	10	—
	Right	27	59	11	14	1	—
	Bilateral	—	5	2	2	—	—
	Unknown	119	167	16	62	6	6
Localization	Frontal	64	82	5	9	—	—
	Fronto-temporal	81	10	8	6	1	—
	Fronto-central	—	—	2	6	2	—
	Fronto-centro-temporal	3	1	—	2	—	—
	Fronto-centro-parietal	12	1	—	1	—	—
	Temporal	60	176	17	6	5	—
	Temporo-occipital	—	1	—	4	—	—
	Temporo-parietal	—	3	1	—	—	—
	Temporo-parieto-occipital	—	—	—	7	—	—
	Central	11	—	—	—	—	—
	Centro-parietal	6	10	1	—	—	—
	Centro-temporal	1	—	—	—	—	—
	Centro-temporo-parietal	6	—	—	1	—	—
	Occipital	5	6	—	3	3	—
	Occipito-parietal	—	1	—	—	—	—
	Parietal	—	—	1	—	—	—
	Parieto-temporal	2	—	1	—	—	—
	Insula	13	—	1	4	—	—
	Unknown	52	100	18	49	6	6

Table 1. Number of seizures in the dataset for each type, lateralization and localization (FA- focal aware; FIA- focal impaired awareness; FBTC- focal-to-bilateral tonic clonic).

‘recordingDuration’ respectively), onset of the event in seconds (in the ‘onset’ field), duration of the event in seconds (in the ‘duration’ field), type of event (in the ‘eventType’ field) and, in the case of a seizure event, the lateralization (hemisphere in which the seizure onset is observed), localization (cerebral lobe of the seizure onset) and vigilance (state of the patient during the seizure) are included as well.

Technical Validation

Data quality. All participants were monitored during the recordings, with daily checks on the wearable device placement and its impedance. Epileptologists and clinical neurophysiologists from each center evaluated the vEEG data of every participant to annotate seizure occurrences. The annotations were based on video records of the patients during monitoring and the full-scalp EEG data recorded with the EMU equipment. The annotations and the wearable data of this dataset were carefully aligned with the full-scalp EEG data. The sampling frequency of the wearable data was matched to the full-scalp EEG data (256 Hz for the bte-EEG, ECG and EMG and 25 Hz for the ACC and GYR). Both devices have an internal clock, which record the time and date of the recording. However, due to differences in hardware, the clock between the vEEG and SD can become misaligned. Additionally, the sampling frequency of the SD is not exact, causing time-wrapping and increasing delays between data recorded with the wearable device and vEEG. The time alignment and delays/interferences were corrected via cross-correlation between the bte-EEG and the full-scalp EEG data. For this alignment, the full-scalp EEG data was firstly converted into a montage similar to the bte-EEG data using the T3, T4, T5 and T6 electrodes. By calculating the cross-correlation between one channel of the SD and the corresponding channel on the vEEG (for example, the left same-side channel and the T3-T5 channel) within a window of 5 minutes, we can obtain the exact misalignment time accurately (in the order of milliseconds). We repeat this process throughout the recording and adjust the time-wrapping effects by stretching/compressing the SD data in order to match the delays obtained with the cross-correlation. The seizures recorded, even if not visible in the data recorded with the SD device, are true seizures experienced by the patients. During data alignment, files in which the wearable data was completely corrupted (the recording contained only flat lines), were removed from the dataset.

Seizure detection. The main objective of the dataset is to promote the development of automated seizure detection frameworks based on wearable data for continuous patient monitoring outside of the EMU. In this work, we implemented two different seizure detection methodologies, one based on a feature-based ML architecture and another using a deep learning framework. These serve as baseline methods for future seizure detection works with the dataset. In order to standardize the comparison between methods, we proposed a division between training and validation data. The training set contains 80% of the data and the validation set 20%. The division was made with efforts to keep the proportions of the number of seizures equal to the proportion of the amount of data in each set, as well as the number of patients from each center. Additionally, we attempted to

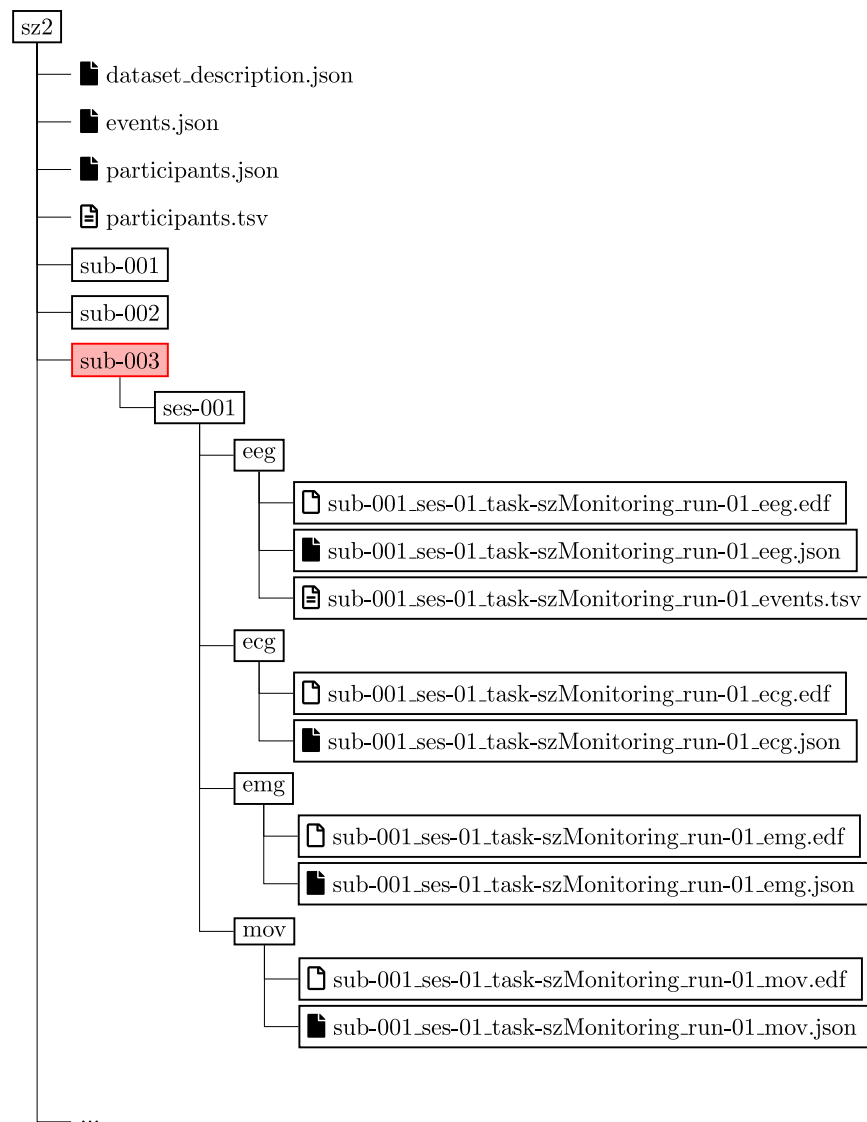


Fig. 3 File hierarchy of the repository.



Fig. 4 Number of seizures per main type in the training and validation sets (FA- focal aware; FIA- focal impaired awareness; FBTC- focal-to-bilateral tonic clonic).

keep the proportion of each seizure type equal within sets. The training set corresponds to the data from patients sub-001 to sub-096 and the validation set from patients sub-097 to sub-125 from the repository. The patient numbering is unrelated to any specific order. In total, 702 seizures were included in the training set and 181 in the validation set. The graph in Fig. 4 illustrates the distribution of seizure types by set.

The feature-based method was based on a previous study¹⁸. In this work, a Support Vector Machine (SVM) model was used to detect seizures in bte-EEG recorded with hospital equipment. This method involves an initial

	Sens	FA/h	AUROC	AUPR	AUSF
SVM	71.1	11.0	0.6995	0.0810	0.7229
ChronoNet	84.2	100.5	0.7921	0.1042	0.7953

Table 2. Performance of the SVM and ChronoNet models on the validation set.

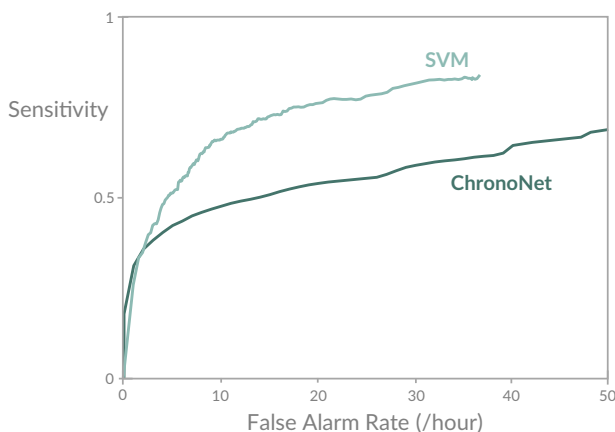


Fig. 5 Sensitivity - false alarm rate per hour curve of the two automated seizure detection methods.

pre-processing of the data with standard EEG filtering (1-25 Hz band-pass Butterworth filter) and data segmentation, creating 2-second EEG windows with a 50% overlap used as input to the model. Segments were discarded if their root mean square amplitude was higher than $150 \mu V$ or lower than $13 \mu V$. A major difference between the work of Vandecasteele *et al.*¹⁸ and this manuscript is the number of channels. The previously used model was developed on 3-channel bte-EEG (two same-side channels, left and right, and one cross-head channel). The method was adapted to receive as input 2-channel bte-EEG, reducing the number of features to 42, since the same-side power asymmetry features are not relevant with this setup. Another difference is the label pruning. Previously, only the seizure data that was clearly visible on the wearable modality was included. With the SeizelT2 data, all segments annotated as 'seizure' were included in the model's training. In order to balance the two classes, the majority class (background EEG) was undersampled to match the number of 'seizure' segments multiplied by a factor of five. The feature-based framework was implemented in MATLAB2019b, with the use of the 'fitcsvm' function, using a radial basis function as a kernel.

The second method is based on a DL architecture, the ChronoNet, that combines both convolutional and recurrent layers³³. The architecture was developed initially for abnormal EEG classification. More recently, it was adapted for seizure detection on bte-EEG³⁴. Similar to the previous method, the input to the model is 2-second bte-EEG segments with 75% overlap for the seizure data and 50% for the background EEG. The pre-processing includes resampling the data to 250 Hz and three Butterworth filters (0.5 Hz high-pass, 60 Hz low-pass and 50 Hz notch filters). The training data is balanced in the same way as the feature-based method, choosing randomly five times the number of seizure segments for the background data segments.

Both models were trained with the training set and evaluated with the validation set defined previously. The evaluation was done with both the traditional epoch-based and the any-overlap methods³⁵. Before evaluation, the model's classification probabilities went through a post-processing procedure. Firstly the segments with a root mean square amplitude below $13 \mu V$ and above $150 \mu V$ are discarded as potential seizure alarms. Furthermore, a positive seizure alarm was kept if, within a window of 10 seconds, there were at least 8 segments of 1 second classified as seizure. The sensitivity and the false alarm rate per hour were used to report the performance of the models. In this work, the area under the receiver operating characteristic curve (AUROC), the area under the precision-recall curve (AUPR) and the area under the sensitivity-normalized false alarm rate curve (AUSF) are also used as comparative metrics. The AUROC and AUPR were derived from metrics calculated using the traditional epoch-based method for computing sensitivity/recall, specificity and precision. The metrics values are presented in Table 2, as well as the sensitivity and the false-alarm rate calculated using a prediction threshold of 0.5. The sensitivity-false alarm rate per hour curve is presented in Fig. 5. The figure displays the sensitivity values and correspondent false alarm rate per hour with varying threshold values for the predicted probabilities. Despite the ChronoNet method surpassing the SVM in all metrics associated to the area under the curves, the maximum sensitivity is lower and the trade-off between sensitivity and false alarm rate of the SVM is more suitable. In a clinical scenario, considering the use of automated models as an assistant tool for annotators, prioritizing sensitivity is more significant, since capturing all seizures leads to a better clinical evaluation of the patients. However, having a significantly high false alarm rate is not desirable.

The performance of the models presented in this study is still sub par to other methods published in the literature^{18,21}. The methods used for seizure detection were developed and validated in the initial iteration of the SeizelT project, where hospital equipment was used to record patients. The wearable device introduces

additional challenges for automated seizure detection frameworks such as added undesirable noise, lower data quality and decreased measurement reliability since the SD can be prone to technical errors and the measurement setup is not standardized. The location and lateralization of the seizures can affect the performance of the models since seizures where the onset is further from the bte-EEG electrodes location are harder to be captured in the data. Other EEG recording methods, such as implantable sub-scalp electrodes, offer a viable solution to concerns regarding data quality and practicality³⁶. These devices can enable more accurate seizure quantification and support additional applications, such as precise seizure focus localization and seizure forecasting³⁷. However, this approach is inherently invasive and requires surgical implantation of the electrodes. Research on the adverse effects of this technique is limited, though some studies report potential complications such as infections, mild headaches, and skin-related symptoms³⁷. Wearable devices overcome the need for surgical implants and are cheaper methods to monitor patients long-term. It can be argued that such devices can be stigmatizing and impractical for patients, where the bulkiness and appearance of the devices causes social barriers in the patients' lives³⁸. A longitudinal study using the SD device in patients with focal epilepsy outside the hospital setting demonstrated promising results in terms of patient acceptance and clinical usability²⁸. The willingness to use the device was largely driven by a desire to improve diagnostic yield and to enable better monitoring of seizure progression. Only minor limitations were reported, primarily related to adverse effects caused by the device. One of the major drawbacks was the limited performance of the automated methods to detect seizures. Additionally, the recording environment in which the data were obtained is the hospital. The conditions of the intended daily-life scenario in which the wearable device is intended to be used is not mimicked in the current dataset. The Sensor Dot was used by a selection of patients outside of the hospital^{27,28}, however the data are not currently public. The presented dataset allows further development of these methods to implement such frameworks in clinical practice, and ease the burden on patients, care-givers and clinicians.

Usage Notes

The dataset can be loaded and manipulated using the pipelines shared in our git repository (<https://github.com/biomedepi/seizeit2>). We include a Python data loader developed in Python 3.10.4, using the pyEDFlib v0.1.38 and pandas v2.2.3 packages.

Code availability

All codes used to reproduce the work presented in this manuscript, including data pre-processing, model training and validation, can be accessed at <https://github.com/biomedepi/seizeit2>. The models were implemented with TensorFlow v2.10.0.

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References

- Steinmetz, J. D. *et al.* Global, regional, and national burden of disorders affecting the nervous system, 1990–2021: a systematic analysis for the Global Burden of Disease Study 2021. *The Lancet Neurology* **23**, 344–381, [https://doi.org/10.1016/s1474-4422\(24\)00038-3](https://doi.org/10.1016/s1474-4422(24)00038-3) (2024).
- Fisher, R. S. *et al.* ILAE Official Report: A practical clinical definition of epilepsy. *Epilepsia* **55**, 475–482, <https://doi.org/10.1111/epi.12550> (2014).
- Fisher, R. S. *et al.* Epileptic seizures and Epilepsy: Definitions proposed by the International League against Epilepsy (ILAE) and the International Bureau for Epilepsy (IBE). *Epilepsia* **46**, 470–472, <https://doi.org/10.1111/j.0013-9580.2005.66104.x> (2005).
- Chauvel, P. & McGonigal, A. Emergence of semiology in epileptic seizures. *Epilepsy & Behavior* **38**, 94–103, <https://doi.org/10.1016/j.yebeh.2013.12.003> (2014).
- Fisher, R. S. *et al.* Operational classification of seizure types by the International League Against Epilepsy: Position Paper of the ILAE Commission for Classification and Terminology. *Epilepsia* **58**, 522–530, <https://doi.org/10.1111/epi.13670> (2017).
- Bruno, E. *et al.* Wearable technology in epilepsy: The views of patients, caregivers, and healthcare professionals. *Epilepsy & Behavior* **85**, 141–149, <https://doi.org/10.1016/j.yebeh.2018.05.044> (2018).
- Thijs, R. D., Ryvlin, P. & Surges, R. Autonomic manifestations of epilepsy: emerging pathways to sudden death? *Nature Reviews Neurology* **17**, 774–788, <https://doi.org/10.1038/s41582-021-00574-w> (2021).
- Salas-Puig, X., Iniesta, M., Abaira, L. & Puig, J. Accidental injuries in patients with generalized tonic–clonic seizures. A multicenter, observational, cross-sectional study (QUIN-GTC study). *Epilepsy & Behavior* **92**, 135–139, <https://doi.org/10.1016/j.yebeh.2018.10.043> (2019).
- Hoppe, C., Poepel, A. & Elger, C. E. Epilepsy: Accuracy of Patient Seizure Counts. *Archives of Neurology* **64**, 1595, <https://doi.org/10.1001/archneur.64.11.1595> (2007).
- Swinen, L. *et al.* Accurate detection of typical absence seizures in adults and children using a two-channel electroencephalographic wearable behind the ears. *Epilepsia* **62**, 2741–2752 (2021).
- Rosenow, F. Presurgical evaluation of epilepsy. *Brain* **124**, 1683–1700, <https://doi.org/10.1093/brain/124.9.1683> (2001).
- Moseley, B. D., Dewar, S., Haneef, Z. & Stern, J. M. How long is long enough? The utility of prolonged inpatient video EEG monitoring. *Epilepsy Research* **109**, 9–12, <https://doi.org/10.1016/j.eplepsyres.2014.10.011> (2014).
- Wang, E. T. *et al.* Seizure count forecasting to aid diagnostic testing in epilepsy. *Epilepsia* **63**, 3156–3167, <https://doi.org/10.1111/epi.17415> (2022).
- Dunn, J., Runge, R. & Snyder, M. Wearables and the medical revolution. *Personalized Medicine* **15**, 429–448, <https://doi.org/10.2217/pme-2018-0044> (2018).
- Byteflies. Online; accessed 24 September 2014.
- Chatzichristos, C., Claro Bhagubai, M., De Vos, M. & Van Paesschen, W. SeizeIT1, <https://doi.org/10.48804/P5Q00J> (2023).
- Gu, Y. *et al.* Comparison between Scalp EEG and Behind-the-Ear EEG for Development of a Wearable Seizure Detection System for Patients with Focal Epilepsy. *Sensors* **18**, 29, <https://doi.org/10.3390/s18010029> (2017).
- Vandecasteele, K. *et al.* Visual seizure annotation and automated seizure detection using behind-the-ear electroencephalographic channels. *Epilepsia* **61**, 766–775, <https://doi.org/10.1111/epi.16470> (2020).
- Leijten, F. S. S. Multimodal seizure detection: A review. *Epilepsia* **59**, 42–47, <https://doi.org/10.1111/epi.14047> (2018).
- Vandecasteele, K. *et al.* The power of ECG in multimodal patient-specific seizure monitoring: Added value to an EEG-based detector using limited channels. *Epilepsia* **62**, 2333–2343, <https://doi.org/10.1111/epi.16990> (2021).

21. Bhagubai, M. *et al.* The power of ECG in Semi-Automated seizure detection in addition to Two-Channel Behind-the-Ear EEG. *Bioengineering* **10**, 491, <https://doi.org/10.3390/bioengineering10040491> (2023).
22. Siddiqui, M. K., Morales-Menendez, R., Huang, X. & Hussain, N. A review of epileptic seizure detection using machine learning classifiers. *Brain informatics* **7**, 5 (2020).
23. Craik, A., He, Y. & Contreras-Vidal, J. L. Deep learning for electroencephalogram (eeg) classification tasks: a review. *Journal of neural engineering* **16**, 031001 (2019).
24. De Cooman, T. *et al.* Online Automated Seizure detection in temporal lobe Epilepsy patients using Single-lead ECG. *International Journal of Neural Systems* **27**, 1750022, <https://doi.org/10.1142/s0129065717500228> (2017).
25. Beniczky, S., Conradsen, I., Henning, O., Fabricius, M. & Wolf, P. Automated real-time detection of tonic-clonic seizures using a wearable EMG device. *Neurology* **90**, <https://doi.org/10.1212/wnl.0000000000004893> (2018).
26. Chatzichristos, C. *et al.* Multimodal detection of typical absence seizures in home environment with wearable electrodes. *Frontiers in Signal Processing* **2**, 1014700 (2022).
27. Swinnen, L. *et al.* Home recording of 3-hz spike-wave discharges in adults with absence epilepsy using the wearable sensor dot. *Epilepsia* **65**, 378–388 (2024).
28. Macea, J., Bhagubai, M., Broux, V., De Vos, M. & Van Paesschen, W. In-hospital and home-based long-term monitoring of focal epilepsy with a wearable electroencephalographic device: Diagnostic yield and user experience. *Epilepsia* **64**, 937–950 (2023).
29. Handa, P., Mathur, M. & Goel, N. EEG datasets in machine learning applications of epilepsy diagnosis and seizure Detection. *SN Computer Science* **4**, <https://doi.org/10.1007/s42979-023-01958-z> (2023).
30. Dan, J. *et al.* SzCORE: Seizure Community Open-Source Research Evaluation framework for the validation of electroencephalography-based automated seizure detection algorithms. *Epilepsia* <https://doi.org/10.1111/epi.18113> (2024).
31. Bhagubai, M. *et al.* SeizeIT2, <https://doi.org/10.18112/openneuro.ds005873.v1.1.0> (2025).
32. Gorgolewski, K. J. *et al.* The brain imaging data structure, a format for organizing and describing outputs of neuroimaging experiments. *Scientific Data* **3**, <https://doi.org/10.1038/sdata.2016.44> (2016).
33. Roy, S., Kiral-Kornek, I. & Harrer, S. Chrononet: A deep recurrent neural network for abnormal eeg identification (2018). 1802.00308.
34. Bhagubai, M. *et al.* Towards automated seizure detection with wearable eeg – grand challenge. *IEEE Open Journal of Signal Processing* **5**, 717–724, <https://doi.org/10.1109/OJSP.2024.3378604> (2024).
35. Shah, V., Golmohammadi, M., Obeid, I. & Picone, J. *Objective Evaluation Metrics for Automatic Classification of EEG Events* (pp. 223–255. Springer International Publishing, Cham, 2021).
36. Duun-Henriksen, J. *et al.* A new era in electroencephalographic monitoring? Subscalp devices for ultra-long-term recordings. *Epilepsia* **61**, 1805–1817, <https://doi.org/10.1111/epi.16630> (2020).
37. Haneef, Z. *et al.* Sub-scalp electroencephalography: A next-generation technique to study human neurophysiology. *Clinical Neurophysiology* **141**, 77–87, <https://doi.org/10.1016/j.clinph.2022.07.003> (2022).
38. Olsen, L. S., Nielsen, J. M., Simonj, C., Kjær, T. W. & Beck, M. Wearables in real life: A qualitative study of experiences of people with epilepsy who use home seizure monitoring devices. *Epilepsy & Behavior* **125**, 108398, <https://doi.org/10.1016/j.yebeh.2021.108398> (2021).

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Competing interests

The authors declare no competing interests.

Additional information

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